Baimam Boukar Jean Jacques

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Research Interests

Recent master's graduate from Carnegie Mellon University working on mechanistic interpretability, interpretability generalization, and the science of evaluations for advanced language models. I study when activation-based monitors and causal interventions survive changes in model family, task, layer, and data distribution, with a particular focus on deception detection. My goal is to build rigorous, reproducible evidence about where white-box safety methods generalize, where they fail, and what those boundaries imply for reliable AI oversight.

Mechanistic interpretability | Interpretability generalization | Deception monitoring | Science of evals

Education

MSc. in Information Technology — Applied Machine Learning — Kigali, Rwanda
Carnegie Mellon University Africa — Aug 2024 – May 2026

BSc. in Computer Science — Software Engineering and Cloud Computing — Yaoundé, Cameroon
Information, Communication and Technology University — Oct 2020 – Jun 2024

Experience

Research Assistant, Mechanistic Interpretability — Toronto, Remote
Jinesis AI Lab at the University of Toronto — Jun 2026 – Present

- Selected high-impact, "load-bearing" mechanistic-interpretability studies for replication using scientific influence, downstream reliance, methodological vulnerability, and reproducibility as selection criteria.
- Audited published probing and steering methods, reproduced selected baselines, and documented failed or incomplete replications while evaluating generalization across models and tasks.
- Built reproducible GPU pipelines for activation extraction, probe training, evaluation, and artifact versioning across Vast.ai, Modal, and Hugging Face.
- Identified baseline replication failures, missing artifacts, and cross-model degradation; developed a taxonomy linking failures to architecture, layer selection, probe design, and data distribution.

Graduate Teaching Assistant — Introduction to Deep Learning — Kigali, Rwanda / Remote
Carnegie Mellon University — Jan 2026 – May 2026

- Led technical tutorials, coordinated course logistics, and graded assignments and projects for CMU's Introduction to Deep Learning course.
- Supported students on LSTMs, GRUs, Transformers, Vision Transformers, GANs, fine-tuning, and RLHF.

Research Scientist Intern — Nairobi, Kenya
IBM Research — Jun 2025 – Sep 2025

- Co-designed and built **IBM DeepScanner**, a PyTorch research tool using non-parametric subset scanning to detect anomalous patterns such as hallucinations and toxicity inside LLM activations.
- Architected a modular workflow for activation and embedding extraction, model analysis, and evaluation across local and distributed IBM GPU clusters, making experiments three times faster.
- Ran end-to-end evaluations across Granite, Llama, Gemma, and Mistral; the work contributed to an AAAI workshop paper and a granted patent.

Graduate Research Assistant — Kigali, Rwanda
Carnegie Mellon University Africa — CyLab Africa — Jan 2025 – May 2025

- Built and maintained research software for Digital Public Goods projects and conducted bi-weekly code reviews focused on reproducibility, code quality, and Open Science practices.

AI Safety Engagement

- **BlueDot Impact — Technical AI Safety Course.** Completed intensive training on emergent capabilities, model evaluations, frontier-lab safety policies, and methods for training safer systems.
- **BlueDot — Technical AI Safety Project Sprint.** Selected for a mentored mechanistic-interpretability sprint and developed a mechanistic audit of deception monitors using probing, causal patching, and steering.
- **Independent replication and audit.** Reproduced Apollo's deception probes, LIARS' Bench, and Matryoshka Sparse Autoencoders, reporting successful baselines alongside failures under model and distribution shift.

Selected Projects

Does Sensing Deception Predict Where to Steer? writeup code Mar 2026 – Present

linear probing | activation steering | causal evaluation | deception

- Built a model-agnostic PyTorch and Hugging Face pipeline for layer-wise activation extraction, L2-regularized probing, mean-ablation patching, and steering across Llama-3.1-8B, OLMo-2-13B, Qwen2.5-32B, and Gemma-2-27B.
- Evaluated controlled on-policy deception scenarios and LIARS' Bench; found that near-perfect probe AUROC did not reliably identify effective intervention layers or deception-specific directions.
- Introduced selection-regret evaluation and norm-matched controls to distinguish useful intervention-site selection from generic activation disruption.

Replication of Matryoshka Sparse Autoencoders writeup code May 2026 – Jun 2026

sparse autoencoders | replication | feature absorption | Modal / H100

- Reproduced the paper's feature-absorption result: the Matryoshka SAE recovered all 20 hierarchical ground-truth features, compared with 11 of 20 for a vanilla SAE despite its lower reconstruction error.
- Rebuilt the LLM-scale pipeline on a TinyStories transformer, cached 30M layer-3 residual activations, and trained matched-sparsity SAEs on three H100 GPUs; published the code and a public LessWrong walkthrough.

Autoregressive Language Modeling with a Causal Transformer Decoder code Nov 2025 – Jan 2026

transformers | PyTorch | HPC | decoding

- Implemented autoregressive and encoder–decoder Transformers from scratch, including multi-head attention, causal masking, positional encodings, and training on HPC clusters.
- Built greedy and beam-search decoders and evaluation pipelines for perplexity and character error rate on LibriSpeech and WSJ.

Selected Publications

Baimam Boukar Jean Jacques (2026). *How Matryoshka Sparse AutoEncoders Recover Feature Hierarchies That Vanilla SAEs Lose.* LessWrong technical article. link

Allassan A. Nken, **Baimam Boukar**, Miriam Rateike, Celia Cintas, Skyler Speakman (2026). *Retrieval with Multiple Query Vectors Through Anomalous Pattern Detection.* AAAI 2026 Workshop on New Frontiers in Information Retrieval. link

Patents

Retrieving Database Vectors From Multiple Query Vectors (2026) Granted

Skills

Programming: Python (PyTorch, NumPy, scikit-learn), C++, Git, Linux
Interpretability and evaluation: Hugging Face, TransformerLens, NNsight, activation hooks, probing, sparse autoencoders, steering, ablations, multi-seed evaluation
Infrastructure and languages: Vast.ai, Modal, IBM GPU clusters, SLURM/HPC, Weights & Biases; English (fluent), French (native)